

DIHWorld Audio

Vocal / Mic / Source Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 6 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-10-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\Phi/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

Silver reference / Gold color / WPR adaptive ribbon

SILVER is the M50-referenced neutral comparison stage. Use it to establish an honest, level-matched baseline for vocal, acoustic, and monitoring decisions: truth, air, articulation, transient precision, and spatial reveal. It is a reference voicing, not an automatic correction or universal flat-response promise.

GOLD uses the M50-referenced input followed by deliberate color. Use it only when the arrangement benefits from harmonic body, presence, soul, weight, or dimensional lift. **GOLD** is not a substitute reference and **SILVER** is not a downstream color stage.

WPR is the pressure-responsive ribbon stage. It adapts Damping, Excursion, and Ribbon Guard to the incoming pressure envelope before the ribbon/transformer path. **WPR** is model- and price-agnostic; exact microphone correction requires measured calibration data.

Decision order: gain-match the dry source and **SILVER** first; compare **GOLD** at the same level when color is wanted; use **WPR** when pressure-responsive ribbon behavior is the intended source stage. Do not prescribe a serial **GOLD**-to-**SILVER** chain as a default.

GOLD

RELEASE	FAMILY	PLACEMENT
v1.0.1	Vocal / Mic / Source	Insert Safe

M50-referenced input response followed by deliberate harmonic color for body, presence, soul, weight, and dimensional lift. GOLD is a creative color stage, not an automatic correction or neutral reference.

Recommended use: Creative mic-color stage after a level-matched reference comparison; use only when deliberate harmonic character is wanted.

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
sourcearray	Source / Mic Array	Factory
AG_AIR	Ag Air	Factory
CU_BODY	Cu Body	Factory
FE_GLUE	Fe Weight	Factory
AU_DRIVE	Au Presence	Factory
PT_AUTHORITY	Pt Authority	Factory
NI_FOCUS	Ni Focus	Factory

Factory starting points

Canon Open • Source True • Root Anchor • Earth Hold • Body Open • Spirit Lift • Sacred Push • Full Field

SILVER

RELEASE	FAMILY	PLACEMENT
v1.0.1	Vocal / Mic / Source	Insert Safe

M50-referenced neutral source voicing for truthful level and tone comparisons: air, articulation, transient precision, and spatial reveal. SILVER is a reference voicing, not an automatic correction or universal flat-response promise.

Recommended use: Neutral reference stage: establish a level-matched baseline before deciding whether GOLD color is musically useful.

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
AG_TRUTH	Truth Field	Factory
NI_FOCUS	Precision	Factory
CU_GROUND	Ground	Factory
AIR	Air	Factory
TRANSIENT	Transient	Factory
OUTPUT	Output	Factory
INPUT_GAIN	Input	Factory
MIX_BLEND	Mix	Factory

Factory starting points

Silver Prep [Stereo] • Reference Truth [Stereo] • Air Open [Stereo] • Transient Detail [Stereo] • Vocal Reveal [Stereo] • String Clarity [Stereo] • Piano Glass [Stereo] • Room Truth [Stereo] • Source True [Stereo] • Phi Balance [Stereo] • Breath Silver [Stereo] • Precision Field [Stereo] • Expansion Air [Stereo] • Neutral Transfer [Stereo] • Silver Prep [Mono] • Reference Truth [Mono] • Air Open [Mono] • Transient Detail [Mono] • Vocal Reveal [Mono] • String Clarity [Mono] • Piano Glass [Mono] • Room Truth [Mono] • Source True [Mono] • Phi Balance [Mono] • Breath Silver [Mono] • Precision Field [Mono] • Expansion Air [Mono] • Neutral Transfer [Mono] • Silver Prep [Dual Mono] • Reference Truth [Dual Mono] • Air Open [Dual Mono] • Transient Detail [Dual Mono] • Vocal Reveal [Dual Mono] • String Clarity [Dual Mono] • Piano Glass [Dual Mono] • Room Truth [Dual Mono] • Source True [Dual Mono] • Phi Balance [Dual Mono] • Breath Silver [Dual Mono] • Precision Field [Dual Mono] • Expansion Air [Dual Mono] • Neutral Transfer [Dual Mono]

WPR

RELEASE	FAMILY	PLACEMENT
v1.0.1	Vocal / Mic / Source	Insert Safe

Pressure-responsive Phi ribbon channel for any user microphone: a mechanical pressure/excursion stage responds to the incoming signal before the transformer/core model. Ribbon Damping, Ribbon Excursion, and Ribbon Guard shape recovery and transient protection without assuming a microphone price, brand, or hidden flat-response correction.

Recommended use: adaptive ribbon source shaping; use after level-matching the user mic

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
DRIVE	Transformer	Float
ARCHITECTURE	Ribbon Tone	Float
DEPTH	Ribbon Mass	Float
WIDTH	Side Width	Float
FOCUS	Polar Focus	Float
ENERGY	Output Trim	Float
CLARITY	Air Edge	Float
MIX	Ribbon Blend	Float
CONVERGENCE	Proximity	Float
DAMPING	Ribbon Damping	Float
EXCURSION	Ribbon Excursion	Float
RIBBON_GUARD	Ribbon Guard	Choice
IRON_DRIVE	Iron Drive	Float
FLUX	Magnetic Flux	Float
HEADROOM	Core Headroom	Float
PRI_BIAS	Primary Bias	Float
SEC_COLOR	Sec Coloring	Float
LOAD	Winding Load	Float
PERMEABILITY	Permeability	Float
HYSTERESIS	Hysteresis	Float
GEOMETRY	Iron Geometry	Float
gainmatch	Gain Match	Bool
LAMINATION	Lamination Type	Choice
TRANSFORMER_FAMILY	Transformer Family	Choice
CORE_MATERIAL	Core Material	Choice
WINDING_TOPOLOGY	Winding Topology	Choice
CORE_AGE	Core Age	Choice
POLAR_MODE	Polar Pattern	Choice
OUTPUT	Output	Float
channelmode	Channel Mode	Choice

Parameter ID	Display name	Type
dualmonolink	Dual Mono Link	Choice

Factory starting points

WP-R MKVI CANON [Stereo] • VINTAGE VELVET [Stereo] • ROOM AMBIENCE [Stereo] • AIR EDGE [Stereo] • ORCHESTRAL PAIR [Stereo] • BRASS TAMER [Stereo] • VOCAL INTIMACY [Stereo] • BROADCAST WARMTH [Stereo] • DARK SHADOW [Stereo] • IRON DRIVE CORE [Stereo] • SILVER FOIL JENSEN [Stereo] • MAGNETIC CRUNCH [Stereo] • LUNDAHL DEEP OPEN [Stereo] • CINEMAG FAT BOTTOM [Stereo] • MU-METAL SILK [Stereo] • HYSTERESIS HEAT [Stereo] • BIFILAR FOCUS [Stereo] • VINTAGE CORE AGE [Stereo] • FIGURE 8 BLISS [Stereo] • BLUMLEIN CORE [Stereo] • GUITAR CAB FOCUS [Stereo] • DRUM OVERHEAD PHI [Stereo] • ACOUSTIC CLARITY [Stereo] • INFINITE MOMENTUM [Stereo] • RESONANT SPACE [Stereo] • DIH WORLD AUDIO [Stereo] • ULTRA CEILING 369 [Stereo] • WP-R MKVI CANON [Mono] • VINTAGE VELVET [Mono] • ROOM AMBIENCE [Mono] • AIR EDGE [Mono] • ORCHESTRAL PAIR [Mono] • BRASS TAMER [Mono] • VOCAL INTIMACY [Mono] • BROADCAST WARMTH [Mono] • DARK SHADOW [Mono] • IRON DRIVE CORE [Mono] • SILVER FOIL JENSEN [Mono] • MAGNETIC CRUNCH [Mono] • LUNDAHL DEEP OPEN [Mono] • CINEMAG FAT BOTTOM [Mono] • MU-METAL SILK [Mono] • HYSTERESIS HEAT [Mono] • BIFILAR FOCUS [Mono] • VINTAGE CORE AGE [Mono] • FIGURE 8 BLISS [Mono] • BLUMLEIN CORE [Mono] • GUITAR CAB FOCUS [Mono] • DRUM OVERHEAD PHI [Mono] • ACOUSTIC CLARITY [Mono] • INFINITE MOMENTUM [Mono] • RESONANT SPACE [Mono] • DIH WORLD AUDIO [Mono] • ULTRA CEILING 369 [Mono] • WP-R MKVI CANON [Dual Mono] • VINTAGE VELVET [Dual Mono] • ROOM AMBIENCE [Dual Mono] • AIR EDGE [Dual Mono] • ORCHESTRAL PAIR [Dual Mono] • BRASS TAMER [Dual Mono] • VOCAL INTIMACY [Dual Mono] • BROADCAST WARMTH [Dual Mono] • DARK SHADOW [Dual Mono] • IRON DRIVE CORE [Dual Mono] • SILVER FOIL JENSEN [Dual Mono] • MAGNETIC CRUNCH [Dual Mono] • LUNDAHL DEEP OPEN [Dual Mono] • CINEMAG FAT BOTTOM [Dual Mono] • MU-METAL SILK [Dual Mono] • HYSTERESIS HEAT [Dual Mono] • BIFILAR FOCUS [Dual Mono] • VINTAGE CORE AGE [Dual Mono] • FIGURE 8 BLISS [Dual Mono] • BLUMLEIN CORE [Dual Mono] • GUITAR CAB FOCUS [Dual Mono] • DRUM OVERHEAD PHI [Dual Mono] • ACOUSTIC CLARITY [Dual Mono] • INFINITE MOMENTUM [Dual Mono] • RESONANT SPACE [Dual Mono] • DIH WORLD AUDIO [Dual Mono] • ULTRA CEILING 369 [Dual Mono]

Meters and visualizers

PEAK: Protected output peak. Use with Gain Match and bypass to judge level fairly; it is not a gain-reduction meter.

STEREO FIELD: Polar outline plus live mid/side position. Confirm the chosen polar pattern and check mono before widening a return.

AEGIS_NOVA

RELEASE	FAMILY	PLACEMENT
v1.0.1	Vocal / Mic / Source	Insert Safe

Vocal shield for plosive and sibilance control with a restrained drive stage and parallel mix.

Recommended use: vocal/source protection and tone

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
gainmatch	Gain Match	Bool
power	Power	Bool
plosive	Plosive	Float
sibilance	Sibilance	Float
drive	Drive	Float
mix	Mix	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Factory starting points

Vocal Shield [Stereo] • Plosive Deflector [Stereo] • Broadcast Guard [Stereo] • Repair Field [Stereo] • Nova Transparent [Stereo] • Air Lock [Stereo] • Rap Shield [Stereo] • Podcast Field [Stereo] • Full Guard [Stereo] • Bass Vocal Lock [Stereo] • Female Vocal [Stereo] • Singer Songwriter [Stereo] • Breath Detail [Stereo] • Stage Vocal Lock [Stereo] • Vocal Shield [Mono] • Plosive Deflector [Mono] • Broadcast Guard [Mono] • Repair Field [Mono] • Nova Transparent [Mono] • Air Lock [Mono] • Rap Shield [Mono] • Podcast Field [Mono] • Full Guard [Mono] • Bass Vocal Lock [Mono] • Female Vocal [Mono] • Singer Songwriter [Mono] • Breath Detail [Mono] • Stage Vocal Lock [Mono] • Vocal Shield [Dual Mono] • Plosive Deflector [Dual Mono] • Broadcast Guard [Dual Mono] • Repair Field [Dual Mono] • Nova Transparent [Dual Mono] • Air Lock [Dual Mono] • Rap Shield [Dual Mono] • Podcast Field [Dual Mono] • Full Guard [Dual Mono] • Bass Vocal Lock [Dual Mono] • Female Vocal [Dual Mono] • Singer Songwriter [Dual Mono] • Breath Detail [Dual Mono] • Stage Vocal Lock [Dual Mono]

Meters and visualizers

INPUT: Incoming peak activity. Confirms source level reaching the shield.

NOVA: Detected plosive/sibilant activity. More light means stronger shield intervention.

TRIPLER

RELEASE	FAMILY	PLACEMENT
v3.5.7	Vocal / Mic / Source	Insert Safe

Three-voice vocal harmonizer with key, scale, chord, pitch correction, voice gain, pan, fine tune, and Phi field control. The current left/right voices use the repaired high-quality overlap-add pitch path for clean stereo harmonies.

Recommended use: vocal/mic array enhancement

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
DIATONIC	DIATONIC	Referenced
CHORD	CHORD	Referenced
KEY	KEY	Referenced
SCALE	SCALE	Referenced
L_GAIN	L_GAIN	Referenced
R_GAIN	R_GAIN	Referenced
C_GAIN	C_GAIN	Referenced
L_PAN	L_PAN	Referenced
R_PAN	R_PAN	Referenced
BYPASS	BYPASS	Referenced
PITCH_CORRECT	PITCH_CORRECT	Referenced
PC_AMOUNT	PC_AMOUNT	Referenced
FINE_TUNE	FINE_TUNE	Referenced
PHI_FIELD	PHI_FIELD	Referenced
L_ON	L_ON	Referenced
R_ON	R_ON	Referenced
OUT_GAIN	OUT_GAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Factory starting points

Default [Stereo] • Wide Open [Stereo] • Gospel 3rds [Major] [Stereo] • R&B Minor Bed [Stereo] • Soul Dom 7 [Stereo] • Trap Major 7 [Stereo] • Trap Minor 7 [Stereo] • Lo Seventh Bed [Stereo] • Pop Fifth Stack [Stereo] • Streaming Ready [Stereo] • Vinyl Warm [Stereo] • 357 Canon [Full] [Stereo] • 357 Diatonic Major [Stereo] • 357 Diatonic Minor [Stereo] • Default [Mono] • Wide Open [Mono] • Gospel 3rds [Major] [Mono] • R&B Minor Bed [Mono] • Soul Dom 7 [Mono] • Trap Major 7 [Mono] • Trap Minor 7 [Mono] • Lo Seventh Bed [Mono] • Pop Fifth Stack [Mono] • Streaming Ready [Mono] • Vinyl Warm [Mono] • 357 Canon [Full] [Mono] • 357 Diatonic Major [Mono] • 357 Diatonic Minor [Mono] • Default [Dual Mono] • Wide Open [Dual Mono] • Gospel 3rds [Major] [Dual Mono] • R&B Minor Bed [Dual Mono] • Soul Dom 7 [Dual Mono] • Trap Major 7 [Dual Mono] • Trap Minor 7 [Dual Mono] • Lo Seventh Bed [Dual Mono] • Pop Fifth Stack [Dual Mono] • Streaming Ready [Dual Mono] • Vinyl Warm [Dual Mono] • 357 Canon [Full] [Dual Mono] • 357 Diatonic Major [Dual Mono] • 357 Diatonic Minor [Dual Mono]

EchoLocate

RELEASE	FAMILY	PLACEMENT
v1.0.1	Vocal / Mic / Source	Insert Safe

Three-position source-placement processor for spread, drift, microphone gain, proximity, pan, and reflected-space mix.

Recommended use: source localization

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
MODE	MODE	Referenced
SPREAD	SPREAD	Referenced
DRIFT	DRIFT	Referenced
MIX	MIX	Referenced
MIC1_GAIN	MIC1_GAIN	Referenced
MIC2_GAIN	MIC2_GAIN	Referenced
MIC3_GAIN	MIC3_GAIN	Referenced
MIC2_PROX	MIC2_PROX	Referenced
MIC3_PAN	MIC3_PAN	Referenced
OUT_GAIN	OUT_GAIN	Referenced
channelmode	channelmode	Referenced
dualmonolink	dualmonolink	Referenced

Factory starting points

ECHOLOCATE CANON [Stereo] • CLOSE CENTER [Stereo] • WIDE CALL [Stereo] • TRIPLE ROOM [Stereo] • CHEST LOCK [Stereo] • HEAD AIR [Stereo] • WALL RETURN [Stereo] • FOCUS FINDER [Stereo] • PHI PREPARATION [Stereo] • PHI HOME [Stereo] • PHI OPEN [Stereo] • PHI ACTIVE [Stereo] • PHI PUSH [Stereo] • PHI EXPANSION [Stereo] • ECHOLOCATE CANON [Mono] • CLOSE CENTER [Mono] • WIDE CALL [Mono] • TRIPLE ROOM [Mono] • CHEST LOCK [Mono] • HEAD AIR [Mono] • WALL RETURN [Mono] • FOCUS FINDER [Mono] • PHI PREPARATION [Mono] • PHI HOME [Mono] • PHI OPEN [Mono] • PHI ACTIVE [Mono] • PHI PUSH [Mono] • PHI EXPANSION [Mono] • ECHOLOCATE CANON [Dual Mono] • CLOSE CENTER [Dual Mono] • WIDE CALL [Dual Mono] • TRIPLE ROOM [Dual Mono] • CHEST LOCK [Dual Mono] • HEAD AIR [Dual Mono] • WALL RETURN [Dual Mono] • FOCUS FINDER [Dual Mono] • PHI PREPARATION [Dual Mono] • PHI HOME [Dual Mono] • PHI OPEN [Dual Mono] • PHI ACTIVE [Dual Mono] • PHI PUSH [Dual Mono] • PHI EXPANSION [Dual Mono]